

Problem 2: Comparing h2 and h3 against h1

Question: confidence that h2 (and h3) performs worse than h1 on the underlying population, i.e. $P(\text{error}_D(h) > \text{error}_D(h1))$.

Formulae

$$\hat{d} = \text{error}_S(h) - \text{error}_S(h1)$$

$$\hat{\sigma}_d = \sqrt{(e_1(1-e_1)/n + e_h(1-e_h)/n)}, n = 45 \text{ for both (same test set)}$$

Two-sided $N\%$ interval $\hat{d} \pm z_N \hat{\sigma}_d$ leaves $(100-N)/2\%$ in each tail, so the one-sided confidence that $d > 0$ is $(100+N)/2\%$.

h2 vs h1

$$\text{error}_S(h2) = 8.89\% = 4/45 = 0.0889$$

$$\hat{d} = \text{error}_S(h2) - \text{error}_S(h1) = 0.0889 - 0.0667 = 0.0222$$

$$\hat{\sigma}_d = \sqrt{(0.0667 \times 0.9333/45 + 0.0889 \times 0.9111/45)} = \sqrt{(0.001383 + 0.001800)} = 0.0564$$

$$z = \hat{d} / \hat{\sigma}_d = 0.0222 / 0.0564 = 0.39$$

From Table 3: $z = 0.41 \leftrightarrow N = 30\%$, so one-sided confidence $= (100+30)/2 = 65\%$. Our $z = 0.39$ is just below, so confidence $\approx 65\%$ (exact normal CDF: 65.3%).

h3 vs h1

$$\text{error}_S(h3) = 13.33\% = 6/45 = 0.1333$$

$$\hat{d} = \text{error}_S(h3) - \text{error}_S(h1) = 0.1333 - 0.0667 = 0.0667$$

$$\hat{\sigma}_d = \sqrt{(0.0667 \times 0.9333/45 + 0.1333 \times 0.8667/45)} = \sqrt{(0.001383 + 0.002568)} = 0.0629$$

$$z = \hat{d} / \hat{\sigma}_d = 0.0667 / 0.0629 = 1.06$$

From Table 3: $z = 1.00 \leftrightarrow N = 68\% \rightarrow 84\%$; $z = 1.28 \leftrightarrow N = 80\% \rightarrow 90\%$. Our $z = 1.06$ lies just above 1.00, so confidence $\approx 84\text{--}85\%$ (exact normal CDF: 85.6%).

Answer: We can be only about 65% confident that h2 is worse than h1, and about 84-85% confident that h3 is worse than h1. Confidence is higher for h3, because its error gap from h1 (6.67 points) is larger relative to the standard deviation than h2's gap (2.22 points). Neither reaches the conventional 95% level.